
LESSON 4.2: Monitoring and Diagnostic Tools

What this lesson is about

This lesson introduces the procedures to gather information about the most useful Sterling B2B Integrator internal tools into one place and review to monitor and diagnose many performance and system problems with the Sterling B2B Integrator package.

What you should be able to do

After completing this lesson, you should be able to:

- Identify the most commonly used System Logs and their contents.
- Identify the most commonly used System Reports.
- Explain about Message Monitor to check messages.

How you will check your progress

- The progress of the lesson is analyzed based on the successful application of the topics in the scenario and test the exercise independently.

References

Documentation

- Managing System Logs:
https://www.ibm.com/support/knowledgecenter/en/SS3JSW_5.2.0/com.ibm.help.performance_mgmt.doc/SIPM_MngSysLogs.html
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Troubleshoot System Logs

Introduction

Sterling B2B Integrator has many components, including software applications, web servers, and database servers. To monitor the activities of each component, Sterling B2B Integrator generates log files.

Each operations server on a host has its own operations log file. Log files are created in the *install_dir*/logs directory (or, on Windows, the *<install_dir>/logs* directory). To prevent the system log files from taking up needed storage space and memory allocations, Sterling B2B Integrator generates a log file only when the component runs.

New log files are created every 24 hours, and normally at midnight. When the time changes to or from Daylight Saving Time, you must stop and restart Sterling B2B Integrator to create daily log files at the same time as before.

If the daily log file size limit is exceeded, another log file is created. If a pre-determined number of logs is exceeded, old logs are automatically deleted. If the interface links to a deleted log file (which displays a blank page), click a link to a newer log file.



Important

If you are working in a clustered environment, the log information that displays is determined by the node you select from the Select Node list.

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Troubleshoot System Logs

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Log File Naming Conventions

Use the following naming convention for a directory:

- UNIX - install_dir/logs/directory
- Windows - install-dir\logs\directory

In this convention:

- install_dir refers to the name of the installation directory.
- logs refer to the primary log directory.
- directory refers to the subdirectory created when you start Sterling B2B Integrator.

The log file is created in LOGS_daymm_dd_yyyy_hhmmss format.

- LOGS identifies as log directory.
- daymm_dd_yyyy, it starts with the weekday, date in month, date, year format.
- hhmmss is the time in hours, minutes, seconds format.
- directory is the subdirectory created when you start Sterling B2B Integrator. All old log files are moved to the subdirectory for archiving.

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Troubleshoot System Logs

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Log File Naming Conventions

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The naming convention for a log file is name.log.Dyyyymmdd.Thhmmss_#.

- name identifies the type of the log file.
- log is the file name extension, which indicates the type of file.
- Dyyyymmdd is the date in year, month, day format. The D at the beginning of the date means Date.
- Thhmmss is the time in hours, minutes, seconds format. The T at the beginning of the time means Time.
- _# is the increment of the log file. If attempt edto write a log file that exists, then _# is appended to the log file name, allowing to write the new file and save the integrity of the exiting file.



Example

If mylog.D20041101.T092022 exists and you save a new log file under the same name, the new file becomes mylog.D20041101.T092022_2, where the _2 indicates that it is the second in a sequence by using the same file name.



Important

The date and time additions in the naming convention might be part of the log file name. For example, noapp.log does not include the date and time information, but the ui.log.Dyyyymmdd.Thhmmss does include the date and time information.

When you start Sterling B2B Integrator, the install_dir/logs/log files are archived to a time-stamped subdirectory. When the archive is complete, Sterling B2B Integrator writes log files to the install_dir/logs directory again.

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Troubleshoot System Logs

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Viewing Log File Contents



Note

If you are working in a clustered environment, the information that displays is mandated by the node you select from the Select Node list.

To view current log file contents in Sterling B2B Integrator:

1. From the Administration Menu, select Operations > System > Logs.
2. Select the appropriate log file. The log opens.



Note

The Sterling B2B Integrator interface displays only the last 2500 lines of a current log file. To view the entire log, you must have Read permission for the file system where Sterling B2B Integrator is located.

To view old log file contents in Sterling B2B Integrator:

1. In the install_dir/logs/log directory, locate the old log file that you want to view.
2. Open the log file in a text editor in read-only mode. The log contents display.

Analyzing Log File Contents

The contents of a log file can provide information about system activities and problems.

The format for entries that are written to a log file is:

YYYY-MM-DD HH:MM:SS.ss log level 'message code' Scope.Subsystem.Name 'information string'.

In this format:

- YYYY-MM-DD is the date in year, month, day format.
- HH:MM:SS.ss is the time in hour, minutes, seconds format.

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Troubleshoot System Logs

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Analyzing Log File Contents

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- Log level is the condition that is used to determine the type of information that is written to the log. The log level conditions are:
 - FATAL - Collects fatal and critical error information.
 - ERRORDTL - Collects only error conditions, with a detailed description of the error.
 - ERROR - Collects only error conditions such as exceptions and error messages
 - WARN - Collects non-fatal configuration messages.
 - SQLDEBUG - Collects SQL statements that are being run.
 - INFO - Collects basic operational information.
 - TIMER - Collects timing information.
 - COMMTRACE - Collects communication trace information.
 - DEBUG - Collects basic debugging statement including system state and code paths.
 - VERBOSE - Collects extra debugging statements (like XML information) that describe and explain what is happening in the system.
 - ALL - Collects information about all the conditions.

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Troubleshoot System Logs

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Analyzing Log File Contents

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- Message code describes the activity or problem, by using the following format:
 - The first 4 digits specify the scope (like Work-flow, Ops, Util).
 - The next digit specifies the log severity level (default conventions use 1 for error/exception, 2 for Debug messages, 3 for Warnings and 4 for info/All messages).
 - The next 3 digits specify the subsystem (like Work-flow Queue or Work-flow Engine).
 - The last 4 digits specify the error number.
- Scope.Subsystem.Name is a text description of the affected part of Sterling B2B Integrator (like Workflow, Ops, Util); the Sterling B2B Integrator subsystem (like Workflow Queue or Work-flow Engine); and what occurred (Name)
- Information string is a brief description of the activity that occurred.

The following message is an example of this format:

```
[2012-05-30 11:06:55.661] ALL 000440020297
SERVICES.SERVICES_CONTROLLER.INFO_sdi_getName startup: loading
HTTP_SEND_ADAPTER
```

At 11:06:55.661 a.m. on May 30, 2012, Sterling B2B Integrator was started and attempted to load the HTTP Send adapter (HTTP_SEND_ADAPTER). The message also shows the scope (Services); the affected part of Sterling B2B Integrator (Services Controller); what occurred (INFO_sdi_getName); and the error code (0297).

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Troubleshoot System Logs

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Changing Log Settings

You can control the logs in Sterling B2B Integrator by specifying settings for either the global or individual log types.

Changing Log Settings for a Global Log

You can change log settings globally by using the `customer_overrides.properties` file, which prevents customized property file changes from being overridden by updates or patches.

When you change the settings for the global log properties, you must follow the following format for each property:

```
logService.Property=Value
```

where the:

- `logService` identifies the `log.properties` file in the `customer_overrides.properties` file.
- `Property` is the global property of the `log.properties` file that you want to set.
- `Value` is the property setting of the `log.properties` file.



Example

`defaultlog.maxnumlogs=15` – Sets the maximum number of the log type that is specified as the default log (systemlogger by default) to 15.

`uilogger.maxnumlogs=20` – Sets the maximum number of UI type logs to 20.

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Troubleshoot System Logs

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Changing Log Settings

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The following table provides the log type name for each log file name that is defined by default in the log.properties file. If you change the file name, use the original file name from the table to find the log type name.

File name of Log	Log type
alerterlogger.log	alerterlogger
archive.log	archivelogger
Authentication.log	AuthenticationLogger
cdinterop.log	cdinteroplogger
cdinterop_cdjava.log	cdinteropcdjavalogger
ceuiinterop.log	ceulogger
common3splogger.log	common3splogger
corbadapter.log	corbadapter
datastore.log	datastore
delete.log	deletelogger
ebXML.log	ebXMLlogger
EDIINT.log	EDIINTLogger
event.log	event
ftp.log	ftpllogger
ftpclient.log	psftpclientlogger

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Troubleshoot System Logs

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Changing Log Settings

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File name of Log	Log type
http.log	httplogger
httpclient.log	httpclientlogger
jetty.log	jettylogger
lifecycle.log	lifecycleLogger
mailbox.log	mailboxlogger
mgmtdash.log	neo
noapp.log	noapplogger
oftp.log	oftplogger
ops_exe.log	opslogger
Perimeter.log	PSLogger
pipeline.log	pipelinelogger
report.log	reportlogger
resourcemonitor.log	resourcemonitorlogger
mif.log	rnlogger
sap.log	saplogger
schedule.log	schedulelogger
Security.log	SecurityLogger

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Troubleshoot System Logs

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Changing Log Settings

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File name of Log	Log type
servicesctl.log	sclogger
sftpclient.log	sftpclientlogger
sftpserver.log	sftpserverlogger
si_exe.log	sillogger
sql.log	sqllogger
SyncEngine.log	SyncEngineLogger
system.log	systemlogger
system.log	purgellogger
test.log	testlogger
tracking.log	tracking
txtrace.log	txtracellogger
ui.log	uillogger
ui_performance.log	ui_perf_logger
webdav.log	webdavlogger
WebSphereMQSuite.log	wsmqSuiteLogger
webx.log	webxlogger
wf.log	wflogger
wfexception.log	wfexception_logger
wfstatistics.log	wfstatistics

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Troubleshoot System Logs

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Changing Log Settings

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To globally change the property file settings by using customer override property file:

1. In the install_dir/properties directory, locate (or create, if necessary) the customer_overrides.properties file.
2. Open the customer_overrides.properties file in a text editor.
3. Specify the settings for the global log properties with the following format:
`logService.Property=Value`
4. Save and close the customer_overrides.properties file.
5. Stop and restart Sterling B2B Integrator to use the new values.

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Troubleshoot System Logs

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Changing Log Settings

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Changing Log Settings for an Individual Log

For an individual log, you can specify the following information:

- Location of the log file
- The number of lines to the log file to be saved
- The number of details to log

The following table describes the options available to use when you configure the settings for an individual log type:



Note

If you are working in a clustered environment, the information that displays is mandated by the node you select from the Select Node list.

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Troubleshoot System Logs

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Changing Log Settings

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Option	Description
Location	Specifies the absolute path for the log file.
Rollover Interval	If newloggers is set to false, specifies the point at which the oldest lines in the log file are deleted as new lines are created. If newloggers is set to true (default), is the maximum number of lines that are allowed in 50000 lines 100000 lines 150000 lines 200000 lines In this case, the term “lines” refers to logical lines (entries) and not to physical line.
Logging Level	Specifies the amount of detail to log. Choose one of the following values: On – set the logging level to ALL, which includes debugging (creates larger files) Off – set the logging level to ERROR (Default), which only logs errors (creates smaller files) Setting the logging level to ALL could generate an excessive amount of debugging information. You must lower the logging level after you have retrieved the debugging information you need.

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Troubleshoot System Logs

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Changing Log Settings

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To change log settings for an individual log type:

1. From the Administration Menu, select Operations > System > Logs.
2. Click the icon next to the type of log for which you want to change the log settings.
3. On the Log Settings page, specify settings for the options.
4. Click Save.

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Troubleshoot System Logs

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Log File Types

**Note**

In a clustered environment, the information that displays is mandated by the node you select from the Select Node list.

You can access the following log files through the Sterling B2B Integrator interface:

Log Type	Log Name	Description
Central Operations Server		
Operations Security	opsSecurity.log	Used by the security components. Indicates problems with startup, passwords, and passphrases.
Operations Server	ops.log	Used by the operations server.
	opsServer.log	Receives all log messages the operations servers generate at startup.
IBM® Sterling Gentran:Server® for UNIX		
Data Adapter		Logs activities of the Sterling Gentran:Server for UNIX adapter. You cannot turn logging on or off for Sterling Gentran:Server for UNIX data adapter. Note: The Sterling Gentran:Server for UNIX logs are displayed only if you have Sterling B2B Integrator configured for Sterling Gentran:Server for UNIX.

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Troubleshoot System Logs

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Log File Types

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Log Type	Log Name	Description
Application Logs		
Adapter Server	servicesctl.log	Used by the service controller component.
Alerter	alerterlogger.log	Logs notification failures and Alert own errors in the Alert Service. When debug is turned on, it also logs all Alert information, such as all defined alerters and filters information.
Archive	archive.log	Used by the archive components.
Business Process Exceptions	wfexception.log	Tracks the exceptions that occur while a business process is running.
Business Process Execution	wf.log	Captures information specific to running a business process.
Business Process Policy Statistics	wfstatistics.log	Contains work-flow policy statistics generated by the work-flow scheduling policy. The actual content depends on the scheduling policy in place, but basically it contains XML time stamps followed by XML records.
Connect:Direct Server Adapter Protocol Layer	cdinterop_cdjava.log	Used by the Connect:Direct Server adapter.

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Troubleshoot System Logs

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Log File Types

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Log Type	Log Name	Description
Connect:Direct Server and Requester Adapter and Services	cdinterop.log	Used by the Connect:Direct Server and Requester adapter with related services
Connect:Enterprise Server Adapter and Services	ceuinterop.log	Used by the Connect:Enterprise Server adapter.
Corba Adapter	corbaadapter.log	Used by the CORBA components.
Dashboard and Community Management	mgmtdash.log	Used by the Dashboard and Community Management components.
Datastore	datastore.log	Contains messages from any of the datastore components. Commonly used to isolate the cause of an abnormal system process termination.
Delete Resources	delete.log	Logs information about resources that have been deleted from Sterling B2B Integrator.
Document Tracking	tracking.log	Logs document tracking activities.
ebXML Business Process Execution	ebXML.log	Logs ebXML business process execution activities.
EDIINT AS1 and AS2	EDIINT.log	Used by the EDIINT components.

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Troubleshoot System Logs

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Log File Types

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Log Type	Log Name	Description
Event Framework	event.log	Logs event framework activities for events completed in Sterling B2B Integrator.
FTP Client Adapter and Services	ftpclient.log	Used by the FTP Client Adapter and related services.
FTP Server	ftp.log	Used by the FTP server components.
G:Server UNIX Lifecycle	lifecycle.log	Used by the Gentran:Server for UNIX Lifecycle components when loading lifecycle records.
Gentran:Server for UNIX Lifecycle Purge Service	system.log	Used by the Gentran:Server for UNIX Lifecycle purge components when purging lifecycle records.
HTTP Client Adapter and Services	httpclient.log	Used by the HTTP Client Adapter and related services.
HTTP Server Adapter	http.log	Used by the HTTP Server Adapter.
Integrator Administration	ui.log	Used by the Sterling B2B Integrator interface.
Jetty HTTP Server	jetty.log	Used by the Jetty HTTP Server.
Log.ResourceMonitorLog	resourcemonitor.log	Used by the Resource Monitor.

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Troubleshoot System Logs

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Log File Types

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Log Type	Log Name	Description
Mailboxing Subsystem	mailbox.log	Used by the mailbox components in Sterling B2B Integrator.
OFTP Administration	oftp.log	Logs OFTP administration activities.
Perimeter Services	Perimeter.log	Used by the perimeter servers components in Sterling B2B Integrator.
Pipeline	pipeline.log	Used by the pipeline components.
Report	report.log	Used by the reporting components.
RosettaNet Business Process Execution	rnif.log	Used by the RosettaNet TM components.
SAP Adapter Administration	sap.log	Used by the SAP [®] components.
Schedule	schedule.log	Logs scheduling activities.
Security	security.log	Used by the security components. Indicates problems with startup and component licensing.
SFTP Client Adapter and Services	sftpclient.log	Used by the SFTP Client adapter and related services.
SFTP Common Log	common3splogger.log	Logs SFTP security errors.
SFTP Server Adapter	sftpserver.log	Used by the SFTP Server adapter.

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Troubleshoot System Logs

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Log File Types

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Log Type	Log Name	Description
SQL Manager	sql.log	Logs queries sent to the database by the SQL Query service.
SyncEngine Components	SyncEngine.log	Logs synchronization engine activities
System	system.log	Used as a general logging service, typically the default system log.
	noapp.log	Used on an application server independent system as a general activity log.
System Output/Error Redirect	noapp.log	Used to provide additional system log information.
Translation Trace Output	txtrace.log	Used as a logging service that helps with map debugging. This log contains debugging messages that show how the translator traversed the maps definition and how the translator matched each block of data against the map.
User Authentication	Authentication.log	Logs user authentication attempts and activities.
Web Extension	webx.log	Used by the Web Extensions components in Sterling B2B Integrator.

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Troubleshoot System Logs

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Log File Types

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Log Type	Log Name	Description
WebSphereMQ Suite	WebSphereMQSuite.log	Used by the WebSphereMQ Suite Async Receiver adapter and related services.
Windows Service Sterling B2B Integrator log	si_exe.log	Log file created by the Sterling B2B Integrator Windows service.
Windows Service Ops log	ops_exe.log	Log file created by the Opserver Windows service.

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Troubleshoot System Logs

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Log File Types

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The following table describes log files for Sterling B2B Integrator as a Windows service:

Log Name	Description
ScheduleBackup.log	Temporary file that is created when Sterling B2B Integrator Windows service stops.
Backuplogs.log	Temporary file created when Sterling B2B Integrator Windows service stops.
ScheduleStopOps.log	Temporary log file that can be ignored.

The following table describes log files for the DMZ perimeter server:

Log Name	Description
PSLogger.Dyyyymmdd.Thhm mss	Logs perimeter server information for the DMZ perimeter server.
StartupPS.log	Logs startup activities for the DMZ perimeter server.

Exercise 4.2.1: Reading the FTP Server Adapter

Introduction

Alice reads log files that are generated by Sterling B2B Integrator. Alice will know how to troubleshoot communication errors by seeing the difference between log files set with a logging level of All, Communication, and Error.

Instructions

Implementing the FTP Server Adapter

Complete the following steps to implement FTP Server Adapter:

Step 1: From the web browser, access **http://192.168.40.100:9000/dashboard** URL, by specifying **admin** for UserID and **password** for Password.

Step 1: Click **Sign In**.

Step 2: From Sterling B2B Integrator, select **Deployment > Services > Configuration** in Administration Menu.

Step 3: In the Service Configuration page, under the **List** section, in the **Search by Service Type** box, select **FTP Server Adapter** and click **Go!**.

Step 4: Enable the FTP Server Adapter.

Step 5: Click **edit**.

Step 6: In the **FTP Server Adapter: Name** page accept the defaults for the following fields:

- Name
- Description
- Select a group

Step 7: Click **Next**.

Step 8: Using the **FTP Server Adapter: Configuration** page, record the **FTP Listen Port** number _____ and accept the defaults for the following fields and click **Next**.

- FTP Server Listen Port
- Active Data Port Range
- Passive Data Port Range

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Exercise 4.2.1: Reading the FTP Server Adapter

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Instructions

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- Perimeter Server
- Transfer Buffer Size (bytes)
- Minimum Number of Threads
- Maximum Number of Threads

Step 9: In the **FTP Server Adapter: Configuration: Documents** page, accept the defaults and click **Next**.

Step 10: In the **FTP Server Adapter: Configuration** page, accept the defaults for the following fields and click **Next**.

- Should the adapter be restricted to a certain group of users?
- Should the restricted users be assigned a specific range of ports?
- should user start in the directory that matches their user name upon login?

Step 11: In the **FTP Server Adapter: Extractability** page, enter Extractable Count value as 1.

Step 12: In the **FTP Server Adapter: SSL** page, accept the defaults for the following SSL configuration fields and click **Next**.

- SSL
- Key Certificate
- Cipher Strength
- Key Certificate (System Store)
- CA Certificates
- Clear Command Channel

Step 13: Select Enable Service for Business Processes and click **Finish**.

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Exercise 4.2.1: Reading the FTP Server Adapter

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Instructions

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Changing the Logging Level for FTP Server Log

To change the logging level of the FTP Server Log:

1. From the Administration Menu, select **Operations > System > Logs**.
2. Locate **FTP Server** log.
3. Click the pencil icon located to the left of the log.
4. Select **All** from the **Logging Level** from the list and select **Save**.

Testing Login and FTP Access

Complete the following steps to test login and FTP access:

Step 1: From the Windows image, select **Start > Run** and enter **cmd** to open up the command line. The system opens the command-line interface.

Step 2: At the prompt type, cd into **C:\6F89G\Labfiles** Files.

Step 3: Type **ftp** and press Enter.

Step 4: At ftp>, type open and enter the IP address that you are trying to connect with and the port number.

```
ftp> open 192.168.40.100 9032
```

Step 5: Type **admin** for User and **password** for Password, as shown in the screen capture.



```
C:\Users\Administrator>ftp
ftp> open 192.168.40.100 9032
Connected to 192.168.40.100.
220 Server ready for new user.
User (192.168.40.100:(none)): admin
331 Password required for admin.
Password:
230 User admin logged in.
ftp> _
```

Step 6: At the command prompt, type **put C:\6F89G\Labfiles\file1.txt** to add a file to the mailbox system with the put command.

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Exercise 4.2.1: Reading the FTP Server Adapter

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Instructions

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Step 7: To view the files in the repository, enter **dir** to view the result as shown in the screen capture.

```
ftp> put C:\6F89G\Labfiles\file1.txt
200 PORT command successful.
125 Data connection already open; transfer starting.
226 Closing data connection, ASCII transfer complete.
ftp: 103 bytes sent in 0.01Seconds 11.44Kbytes/sec.
ftp> dir
200 PORT command successful.
125 Data connection already open; transfer starting.
drwx----- 1 mailbox group      1024 Sep 21 10:11 DeadLetter
drwx----- 1 mailbox group      1024 Sep 21 10:11 EBClientOrderMetadata
drwx----- 1 mailbox group      1024 Sep 21 10:11 EBICS_DEADLETTER
drwx----- 1 mailbox group      1024 Sep 21 10:11 EDIInboundCollection
drwx----- 1 mailbox group      1024 Sep 21 10:11 EDIInboundExtraction
drwx----- 1 mailbox group      1024 Sep 21 10:11 EDIOutboundCollection
drwx----- 1 mailbox group      1024 Sep 21 10:11 EDIOutboundExtraction
-rw----- 1 message group         98 Sep 21 10:11 file1.txt
drwx----- 1 mailbox group      1024 Sep 21 10:11 veu
226 Closing data connection, ASCII transfer complete.
ftp: 654 bytes received in 0.00Seconds 327.00Kbytes/sec.
ftp>
```

Step 8: At the command prompt, type **get file1.txt** to extract a file from the mailbox system with a **get** command.

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Exercise 4.2.1: Reading the FTP Server Adapter

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Instructions

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Reviewing the FTP Log File

Complete the following steps to review the FTP log file:

Step 1: From the web browser, access **http://192.168.40.100:9000/dashboard** URL, by specifying **admin** for UserID and **password** for Password.

Step 1: Click **Sign In**.

Step 2: From the Administration Menu, select **Operations > Systems > Logs** to view log file area.

Step 3: Select the **ftp.log** Hyperlink.

Step 4: Locate and view the log messages to see the data transfer, as shown in the screen capture.

```
log.FTP Log 09/21/2016 10:05:57 AM - ftp.log
016-09-21 10:05:57.276] ALL FtpServerAdapterImpl.startupAdapter() - Adapter: FTP_SERVER_ADAPTER_FtpServerAdapter_node1 - starting
016-09-21 10:05:57.279] ALL FtpServer.init(P,S) starting initialization.
016-09-21 10:05:57.282] ALL FtpConfig.extractNatConfiguration(P) instanceNATAddress in FtpConfig=null
016-09-21 10:05:57.282] ALL FtpConfig.logConfiguration() server configuration: ftpListenPort=[9032], passiveDataPortCollection=[null]
016-09-21 10:05:57.419] ALL AdapterConcurrencyManager constructed with adapterMax [-1] and userMax [-1]
016-09-21 10:05:57.42] ALL FtpMailboxConfig.logConfiguration() server mailbox configuration: documentStorageType=[1], extractability[1]
```

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Exercise 4.2.1: Reading the FTP Server Adapter

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Instructions

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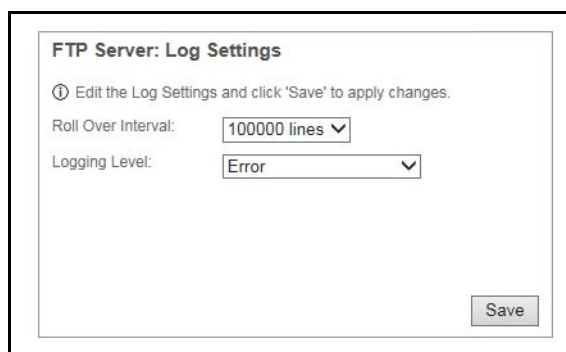
Changing the Logging Level for the FTP Server Log

Complete the following steps to change the logging level of FTP Server Log:

Step 1: From the Administration Menu, select **Operations > System > Logs**.

Step 2: Locate the **FTP Server Log**.

Step 3: Click the pencil icon that is located to the left of the log as shown in the screen capture.



Step 4: Select the **Logging Level** as **Communication Trace** from the drop down-list and click **Save**.



Note

Select the hyperlink for newest ftp.log file.

How this trace is different from All trace. This information is used during classroom discussion.

Exercise 4.2.2: Accessing FTP Server Through Secured Layer

Introduction

Alice generates an error to the Sterling B2B Integrator FTP Server that is configured for SSL without the proper credentials. The errors in the FTP server log with a logging level of communication and then change the logging level to ALL to see additional information.

Instructions

Configuring FTP Server for SSL

Complete the following steps to configure the FTP Server for SSL:

Step 1: From the web browser, access **http://192.168.40.100:9000/dashboard** URL, by specifying **admin** for UserID and **password** for Password.

Step 1: Click **Sign In**.

Step 2: From the **Administration Menu**, select **Deployment > Services > Configuration**.

Step 3: In the **List** section, next to by **Service Type**, select FTP Server Adapter and click **Go!**.

Step 4: From the Services Configuration page, click **edit**.

Step 5: In the Name page, accept the defaults and click **Next**.

Step 6: In the Configuration page, accept the defaults and click Next.

Step 7: In the Extractability page, accept the defaults and click Next.

(Continued on next page)

Exercise 4.2.2: Accessing FTP Server Through Secured Layer

(Continued)

Instructions

....(Continued)

Step 8: In the SSL page, complete the fields in the following table with the parameters given and click **Next**.

Field	Value
SSL	Must
Key Certificate Passphrase	Sterling B2B Integrator
Cipher Strength	STRONG
Key Certificate (System Store)	B2BHttp
CA Certificates	Accept the default
Clear Command Channel	Disallow

Step 9: From the Confirm screen, click **Finish**.

(Continued on next page)

Exercise 4.2.2: Accessing FTP Server Through Secured Layer

(Continued)

Instructions

....(Continued)

Testing FTP Access error without SSL configured for client

Complete the following steps to test login and FTP access:

Step 1: From the Windows image, select **Start > Run** and enter **cmd** to open up the command line. The system opens the command-line interface.

Step 2: At the command prompt, change the directory to **C:\6F89G\Labfiles** directory as.

Step 3: Type **ftp** and press Enter.

Step 4: At ftp>, type open and enter the IP address that you are trying to connect with and the port number.

```
ftp> open 192.168.40.100 9032
```

Step 5: Type **admin** for User and **password** for Password, as shown in the screen capture.



Note

Stop here – the connection must not work because of a secure SSL client is needed. Now, check the FTP log file for errors.

(Continued on next page)

Exercise 4.2.2: Accessing FTP Server Through Secured Layer

(Continued)

Instructions

....(Continued)

Reviewing the FTP Log File

Complete the following steps to review the FTP log file:

Step 1: If log file area is close, From the Administration Menu, select **Operations > Systems > Logs**.

Step 2: Select the **ftp.log** hyperlink

Step 3: Locate and view the log messages msg=[503 to see that the session was rejected.

Step 4: Note the errors listed

Now, change the logging level to All and note the additional information available within the log file.

(Continued on next page)

System Reports

System Reports

The Sterling B2B Integrator Internal System Reports displays the following reports. To access the Sterling B2B Integrator Internal System Reports, from the Administration window select **Operations > Reports > List All**.

Category of Report	Report
adapterjvm	■ AdapterJVMReport
Admin Audit	■ AdminAuditByObjectType ■ AdminAuditByPrincipal
Authentication	■ AuthenticationByEndPoint1 ■ AuthenticationByPrincipal ■ AuthenticationByProtocol
Authorization	■ AuthorizationByPrincipal ■ AuthorizationByProtocol ■ AuthorizationByResourceType
Business Process Definition List	■ BPDefList_ByBPDefStatus ■ BPDefList_ByBPName
Business Process Detail	■ BPDdetail_ByBasicStatus ■ BPDdetail_ByBPName ■ BPDdetail_ByState
DB Stats	■ dbstats - Displays BPs by name, number of runs, and status.
deprecationreport	■ DeprecationReport

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System Reports

(Continued)

System Reports

....(Continued)

Category of Report	Report
EDI Reports	■ EDIOutAck_GroupLevel_ByAckStatus ■ EDIOutAck_GroupLevel_BySenderReceiver ■ EDIOutAck_InterchangeLevel_ByAckStatus ■ EDIOutAck_InterchangeLevel_BySenderReceiver ■ EDITransDetail_GroupLevel_ByEnvelope ■ EDITransDetail_GroupLevel_BySenderReceiver ■ EDITransDetail_InterchangeLevel_ByEnvelope ■ EDITransDetail_InterchangeLevel_BySenderReceiver ■ EDITransDetail_TransacLevel_ByEnvelope ■ EDITransDetail_TransacLevel_BySenderReceiver
Map Lists	■ MapList_ByMapName ■ MapList_ByMapType
SystemDetails Logs	■ systemDetails_Logs
Traffic	■ TrafficByAdapter ■ TrafficByEndPoint1 ■ TrafficByProtocol
Traffic Summary	■ TrafficSummaryByAdapter ■ TrafficSummaryByEndPoint1 ■ TrafficSummaryByProtocol
Translation Service	■ TranslationService

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System Reports

(Continued)

System Reports

....(Continued)

Category of Report	Report
WFFactModel Detail	■ WFFactModelDetail BPAAdvRes MIN ■ WFFactModelDetail BPAAdvWait ■ WFFactModelDetail BPComp MIN ■ WFFactModelDetail BPCont ■ WFFactModelDetail BPContErr ■ WFFactModelDetail BPDeActRes MIN ■ WFFactModelDetail BPDeadline At ■ WFFactModelDetail BPDeadline Chk ■ WFFactModelDetail BPDeadline Ms ■ WFFactModelDetail BPErrComplete ■ WFFactModelDetail BPError ■ WFFactModelDetail BPErrStart ■ WFFactModelDetail BPException ■ WFFactModelDetail BPFTerm

(Continued on next page)

System Reports

(Continued)

System Reports

....(Continued)

Category of Report	Report
WFFactModelDetail	■ WFFactModelDetail BPINT MIN ■ WFFactModelDetail BPREIDBErr ■ WFFactModelDetail BPREsumed MIN ■ WFFactModelDetail BPREsumeErr ■ WFFactModelDetail BPREs MIN ■ WFFactModelDetail BPStarted MIN ■ WFFactModelDetail BPSubStErr ■ WFFactModelDetail BPSubSt MIN ■ WFFactModelDetail BPTerm MIN ■ WFFactModelDetail BPThrowable ■ WFFactModelDetail BPWait ■ WFFactModelDetail CnxtCachWarn ■ WFFactModelDetail DelExpCon MIN ■ WFFactModelDetail RemEventErr ■ WFFactModelDetail RemEvent MIN ■ WFFactModelDetail ServEnded ■ WFFactModelDetail ServError ■ WFFactModelDetail ServStarted

Message Monitor

Introduction

In Sterling B2B Integrator, business processes use messages to communicate with each other. Certain business processes create Produce and Consume messages (asynch communication hand off to the database). The Produce business processes produce messages to be consumed. The Consume business processes wait for messages from the Produce business processes.

Sometimes the hand off between the Produce and Consume business processes does not occur—the produced message has no waiting Consume process, or the waiting Consume process waits without success for the produced message. Use the Message Monitor to track unsuccessful message hand-offs within Sterling B2B Integrator.

Special Considerations

Some special considerations about messages that are monitored within Sterling B2B Integrator:

- The Message Monitor pages of the Sterling B2B Integrator interface show only messages that are associated with a FAILED Produce or Consume activity. Messages that are successfully produced or consumed are not monitored.
- Messages are tied closely to business processes. Search for messages by business process name, type of business process (Produce or Consume), and date and time that the associated business process step invokes the Produce or Consume activity. But messages also have identities separate from business processes. Messages have names, and their names can be used as search input.

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Message Monitor

(Continued)

Special Considerations

- Sometimes Produce and Consume business processes must choose between messages with the same name. For example, two Produce processes that are run at different times; both produce a message that is named OutMsg1. So the two messages in Sterling B2B Integrator named OutMsg1, but with different start date/times. When a Consume process detects the two messages, it consumes the OutMsg1 with the later start date/time. The situation also works in reverse. Two Consume processes request an identically named message (InMsg1). When a Produce process produces InMsg1, the Produce process produces InMsg1 for the Consume process with the earlier start date/time.
- An unsuccessful message remains in Sterling B2B Integrator until an appropriate Produce or Consume business process step completes the message hand off and uses the message.

**Note**

Unsuccessful messages NEVER expire or time out.

Exercise 4.2.3: Check in the AT_produce and AT_consume business process

Introduction

Alice wants to test the consume bpml activity. Alice received two business process from the development team, which he checks in the AT_produce and AT_consume business process to run the AT_produce business process to send a message to the AT_consume business process.

Instructions

AT_produce

Complete the following steps to check in the AT_produce business process.

Step 1: From the web browser, access **http://192.168.40.100:9000/dashboard** URL, by specifying **admin** for UserID and **password** for Password.

Step 1: Click **Sign In**.

Step 2: From the Administration Menu, select **Business Processes > Manager**. The **Business Process Manager** page is displayed.

Step 3: Click **Go!** Next to Process Definition under Create.

Step 4: Type **AT_produce** as **Name**.

Step 5: Select Check in Business Process that is created by the graphical modeling tool and click **Next**.

Step 6: Browse to the **C:\6F89G\Labfiles** directory and select **AT_produce.bp** for the file name.

Step 7: Enter **produce** for the description and click **Next** to continue.

Step 8: Set the persistence level to be **Full** and take the remaining defaults, click **Next** to continue.

Step 9: Do not set a deadline for this process, click **Next** to continue.

Step 10: Take the default for life span, click **Next** to continue.

Step 11: Verify that the business process is enabled and click **Finish**.

Step 12: Click **Return**, the Business Process Manager page is shown

(Continued on next page)

Exercise 4.2.3: Check in the AT_produce and AT_consume business process

(Continued)

Instructions

....(Continued)

AT_Consume

Complete the following steps to check in the consume business process:

- Step 1:** From the Administration Menu, select **Business Processes > Manager**. The **Business Process Manager** page is displayed.
 - Step 2:** Click **Go!** Next to Run Graphical Process Modeler.
 - Step 3:** Enter User ID as **admin** and Password as **password**. The Graphical Process Modeler opens.
 - Step 4:** Click **File** menu and select **Open**.
 - Step 5:** Go to **C:\6F89G\Labfiles** folder and select AT_consume.bp.
 - Step 6:** Click **Consume**. The consume property editor opens.
 - Step 7:** Enter **AT_consume** in the **Participant** text box.
 - Step 8:** Click the **Save** icon and select **Yes** to validate on save.
 - Step 9:** Close the Graphical Process Modeler.
 - Step 10:** In the **Business Process Manager** page, click **Go!** next to Process Definition under Create.
 - Step 11:** Type **AT_consume** as **Name**.
 - Step 12:** Select Check in Business Process that is created by the graphical modeling tool and click **Next**.
 - Step 13:** Browse to the **C:\6F89G\Labfiles** and select **AT_consume.bp** for the file name.
 - Step 14:** Enter **consume** for the description and click **Next** to continue.
 - Step 15:** Set the persistence level to be **Full** and accept the remaining defaults, click **Next** to continue.
 - Step 16:** Do not set a deadline for this process, click **Next** to continue.
 - Step 17:** Take the default for life span, click **Next** to continue.
 - Step 18:** Verify that the business process is enabled and click **Finish**.
-

Exercise 4.2.4: Run the AT_Produce Business Process with InventoryOf11.xml

Instructions

Complete the following steps to execute the process:

- Step 1:** From the Administration Menu, select **Business Processes > Manager**. The **Business Process Manager** page is displayed.
- Step 2:** In the Search text box, type AT_produce and click **Go!**.
- Step 3:** Click the **execution manager** icon for the process.
- Step 4:** Click the **execute** icon.
- Step 5:** Browse to C:\6F89G\Labfiles and select **InventoryOf11.xml**.
- Step 6:** Click **Open** and then **Go!**. Observe the separate Execute Business Process window while the business process runs.
- Step 7:** Click the info of Produce Service to view the Consume process initiated message with the instance ID as 21694. In your lab, the Instance ID would be different.

Name: AT_produce Instance ID: 21691 User: admin

Completed
Status: Success

Step	Service	Status	Advanced Status	Started	Ended	Status Report	Document	Instance Data
3	Produce Service	Success	None	09/22/2016 2:36:57 AM EDT	09/22/2016 2:36:57 AM EDT	Info	Info	Info
2	Spawn Service	Success	None	09/22/2016 2:36:57 AM EDT	09/22/2016 2:36:57 AM EDT	Info	Info	Info
1	XML Encoder Service	Success	None	09/22/2016 2:36:57 AM EDT	09/22/2016 2:36:57 AM EDT	None	Info	Info
0	INITIATING_CONTEXT from user 'admin'	Success	None	09/22/2016 2:36:57 AM EDT	09/22/2016 2:36:57 AM EDT	None	Info	Info

Administration - Internet Explorer
http://192.168.40.100:9000/ws/Tracker?action=getstatus&next=page.wfctrackerinfo&wfd_id=862&workflow_id=21

Execute Business Process

Name: AT_produce Instance ID: 21691 Service Name: Spawn Service

Status Report:

Initiated Find Message Consumer process, instance ID(s): 21694

- Step 8:** Close the Execute Business Process window when the business process is complete.

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Exercise 4.2.4: Run the AT_Produce Business Process with InventoryOf11.xml

(Continued)

Instructions

....(Continued)

Step 9: From Administration Menu select Business Processes > Monitor > Current Processes to view AT_consume process is executed

Step 10: Click AT_consumer Instance ID to analyze the result, as shown in the screen capture.

Steps 1-2 of 2

Step	Service	Status	Advanced Status	Started	Ended	Status Report	Document	Instance Data
0	Spawn Service	Success	None	09/22/2016 3:36:34 AM	09/22/2016 3:36:34 AM	None	Info	Info
1	Consume Service Subprocess 21820**	Waiting	None	09/22/2016 3:36:34 AM	09/22/2016 3:36:34 AM	Info	Info	Info

* Inline Invocation
** The System cannot display processes run with certain persistence levels, or processes that have been archived, purged or deleted.
Last update on 09/22/2016 3:40:01 AM

Exercise 4.2.5: Create an Unsuccessful Message

Instructions

Complete the following steps to create an unsuccessful message by running AT_consume process:

- Step 1:** From the Administration Menu, select **Business Processes > Manager** from the **Administration Menu**, the Business Process Manager page appears.
- Step 2:** In the Search text box, type **AT_consume** and click **Go!**.
- Step 3:** Click the **execution manager** icon for the process.
- Step 4:** Clear Enable the check box and click **Ok** to confirm.
- Step 5:** Click **Return**.
- Step 6:** Click **Return** to go back to the **Business Process Manager** page.
- Step 7:** In the Search text box, type **AT_produce** and click **Go!**.
- Step 8:** Click the **execution manager** icon for the process.
- Step 9:** Click the **execute** icon.
- Step 10:** Browse to **C:\6F89G\Labfiles** and select **InventoryOf11.xml**.
- Step 11:** Click Open and then **Go!**. Observe the separate Execute Business Process window while the business process runs.
- Step 12:** Close the Execute Business Process window when the business process is complete.
- Step 13:** From the Administration Menu, select **Business Processes > Monitor > Current Processes** to view the result as shown in the screen capture.

Status	ID	Name	State	Started
	23077	At consume	Halted	09/22/2016 9:29:49 AM
	23076	AT produce	Completed	09/22/2016 9:29:49 AM

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Exercise 4.2.5: Create an Unsuccessful Message

(Continued)

Instructions

....(Continued)

Step 14: From the Administration Menu, select **Operations > Message Monitor**. The **Message Monitor** page opens.

Step 15: In the Message Monitor page, under **List**, for both **Alphabetically** and **by choose**, select 'A' and click **Go!**.

Step 16: Verify the information and click the current instance ID of **AT_produce** hyper link to analyze further the status of the process as shown in the screen capture.f

The screenshot displays the Sterling B2B Integrator interface. At the top, there's an 'Action:' dropdown set to 'Restart' and a 'Go!' button. Below this is a table titled 'Steps 1-4 of 4' with columns: Step, Service, Status, Advanced Status, Started, Ended, Status Report, Document, and Instance Data. The table lists four steps, with the third step, 'Produce Service Subprocess 23077', highlighted. Below the table, the 'Sub-Process Detail' section shows information for 'At_consume' (Instance ID: 23077) with a status of 'Error' and 'State: Halted'. The 'Parent Business Process' is 'AT_produce' (Instance ID: 23076). To the right, the 'Execute Business Process' window displays the message: 'At_consume is a disabled Business Process.'

Search Unsuccessful Messages

Introduction

Search for unsuccessful message by using the following search methods:

- Message Name – Display messages whose names contain the specified character or string.
- Start Date From and Start Date To – Display messages for which the related BP step invoked the Produce or Consume activity at the specified start date and time on whether Sterling B2B Integrator has outstanding unsuccessful messages.
 - If Sterling B2B Integrator has no unsuccessful messages, the Start Date From field displays the current system date and time minus 1 minute, and the Start Date To field displays the current system date and time plus 1 minute.
 - Unsuccessful messages in Sterling B2B Integrator, the Start Date From field displays the date and time of the unsuccessful message with the earliest start date and time, minus 1 minute; and the Start Date To field displays the date and time of the unsuccessful message with the start date and time, plus 1 minute.
- Alphabetically or by message type (ALL, Producer, or Consumer) – Display messages initiated by business processes that:
 - Names starting with a particular letter or number that is selected from the Alphabetically list. Default is ALL (all business processes).
 - Are either Produce or Consume processes, or both (ALL).

(Continued on next page)

Search Unsuccessful Messages

(Continued)

Introduction

....(Continued)

The following information is given for each message:

- Type – Type of business process that is associated with the message, either Produce or Consume.
 - ID – ID of the business process that is associated with the message. Click an ID in this column to display the Business Process Details page for the BP.
 - Process Name – Name of the associated Produce or Consume BP. Click a process name in this column to display the BPML of the BP.
 - Searching for Unsuccessful Messages
 - Message Name – Name of the unconsumed message.
 - Start Time – Start date and time of the associated step within the BP.
-

Lesson review

What you have been able to do

After completing this lesson, you should have been able to:

- Identify the most commonly used System Logs and their contents.
 - Identify the most commonly used System Reports.
 - Explain about Message Monitor to check messages.
-